Almouthana Taha Khalfallah

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EDUCATION

Higher National School of Engineers of Tunis – ENSIT — Tunis, Tunisia
Research Master's Degree in Data Science — 2026 – Present
Engineering Degree in Applied Mathematics and Modeling – **1st in Cohort** — 2024 – Present

Preparatory Institute for Engineering Studies of Tunis – IPEIT — Tunis, Tunisia
Preparatory Degree in Physics and Chemistry (PC) — 2022 – 2024

EXPERIENCE

SIGA — Tunis, Tunisia
AI Intern — Jun 2026 – Jul 2026
- Developed HRMind, an AI-powered HR assistant automating leave management, employee requests, and document verification workflows.
- Built the application primarily with Oracle APEX and PL/SQL, integrating OCR and LLM-based document validation.

EXCELLIA Solutions — Tunis, Tunisia
Intern — Jun 2025
- Explored RAG systems and Streamlit for AI application deployment, prototyping ML models using Scikit-Learn, Pandas, and Matplotlib.
- Observed Agile workflows (JIRA, GitLab) and completed practical labs on Azure AI services (AI-900, AI-102).

SELECTED PROJECTS

MindEase | TypeScript, React, Vite, WebExtension (MV3), DeepSeek AI, Q-Learning, KaTeX, OCR | Global Bronze Award - GCD4F 2026
- Designed an on-device Q-learning agent dynamically adapting content presentation to learner interactions.
- Integrated LLM-powered explanations, multimodal content adaptation, OCR, and contextual learning support across web content.

HR Management Agent | Oracle APEX, PL/SQL, Oracle Database, OCR, LLM
- Built an automated HR assistant within Oracle APEX to streamline employee leave applications and document verification.
- Integrated OCR and LLM-based processing with PL/SQL database routines to validate document data against internal policies.

LocMemory | Python, NetworkX, PyTorch, React, FastAPI, Supabase, Ollama
- Designed a local memory system for LLMs combining semantic search with a knowledge graph for structured retrieval.
- Implemented a PPO-based reinforcement learning policy to dynamically prioritize and filter candidate memories at inference.

MindMate | React, FastAPI, Supabase, Python
- Built a full-stack productivity workspace featuring an LLM companion for sentiment analysis and intelligent task extraction.
- Designed a RESTful backend with FastAPI and Supabase for real-time synchronization, session state, and user authentication.
- Deployed the application on Vercel/Render with a CI/CD pipeline via GitHub Actions and a Supabase-backed PostgreSQL database.

Micrograd Reconstruction | Python
- Built an autograd engine implementing backpropagation via reverse-mode automatic differentiation, with a small neural network library on top.

RESEARCH & OPEN SOURCE ACCOMPLISHMENTS

MindEase Research Paper | Co-Author
- Co-authoring a research paper on MindEase's three-layer cognitive and adaptation architecture following its GCD4F 2026 Global Bronze Award.

WnR | Open Source Contribution – github.com/RoderickQiu/wnr (1.1k⋆)
- Added Arabic/French localization with RTL support and a Unicode bidi-isolation fix; merged into a 1.1k-star Electron app.

HONORS & LEADERSHIP

GCD4F 2026 | Global Bronze Award – 3rd Place Worldwide — Beijing, China
- 3rd Place Globally in 9th Global Competition on Design for Futures, hosted by Beijing Normal University in Beijing, China. — Oct 2026

Smart Futures Youth Forum 2026 | International Speaker — Beijing, China
- Delivered a talk on AI-driven adaptive education and participated in an innovation roundtable at Beijing Normal University. — Oct 2026

NERDATA ENSIT CLUB — Sep 2024 – Present
- Organized DataFest 1.0, Inspiring Genius, and The Upside Hack AI-Hackathon 1.0; delivered a Git/GitHub workshop.

Hackathons — Tunis, Tunisia
- CodeCrafters, GCD4FE 8, Seneca Hack 1.0, Geeks Hack 5.0, Hack4Ucar 1.0.

GEO-OLYMPIAD — Mar 2022
- Ranked 4th in the Tunis region and qualified for the national finals of the Tunisian Geography Olympiad.

SKILLS

Programming Languages: Python, C, C++, MATLAB, SQL, PL/SQL, HTML, CSS, JS, TS, WinDev (WLanguage)
Libraries & Data: NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, TensorFlow, PyTorch, OpenCV, HuggingFace, Ollama, NetworkX
Technologies & Tools: Oracle APEX, PostgreSQL, Azure, FastAPI, React, Supabase, Vercel, Netlify, Render, Git, GitHub, GitHub Actions, Streamlit, VS Code, Jupyter Notebook, Linux, Docker, Notion, JIRA
Concepts: Machine Learning, Deep Learning, RAG, Reinforcement Learning, Knowledge Graphs, Agentic Systems, MCP, Tokenization
Languages: Arabic (Native), French (Fluent), English (Fluent – TOEIC 900/990)